Formal Lab Report A

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November 28, 2023

Chem 2081-003

Introduction

The objective of this laboratory experiment is to identify a given unknown compound, specifically referred to as sample A, and subsequently compile a comprehensive lab report. The identification process relies on three distinct sets of data: a nuclear magnetic resonance spectrometer reading (NMR), a gas chromatograph and mass spectrometer reading (GCMS), and an infrared spectroscopy (FTIR) analysis. This formal lab report serves as an opportunity to apply a practical skill widely utilized by chemists. The three spectra readings provided offer valuable insights into the composition and characteristics of the unknown compound. Notably, the NMR data, gas chromatography-mass spectrometry results, and infrared spectroscopy findings collectively contribute to the compound's identification. The significance of this exercise extends beyond the laboratory setting, finding practical applications in various fields. For instance, in the medical field, NMR readings play a crucial role in diagnoses and the identification of both types and quantities of different drugs. Acquiring a foundational understanding of the three spectra readings examined in this report equips individuals with a skill set applicable across diverse career fields.

Body

NMR- Nuclear Magnetic Resonance stands out as the predominant method for characterizing structural aspects in organic chemistry. The four main indicating factors that allow for the reading of NMR spectra are the number of different kinds of protons, integration, chemical shift, and spin-spin coupling. NMR is applicable to any nucleus exhibiting nuclear spin, indicating those with either an odd mass or atomic number. This is due to its ability to unveil the precise positioning and interconnection of atoms. Hydrogen NMR is the most widely utilized form of NMR, primarily because hydrogen, specifically the isotope of hydrogen known as protium, is the most abundant. The mechanism of NMR relies on the magnetic spin exhibited by nuclei to generate results, mainly the C or H nuclei. Nuclei, like magnets with positive charges, align their spin with an external magnetic field. The energy difference induced by radio waves between these two states results in resonance, generating waves and characterizing the protons as being in resonance.

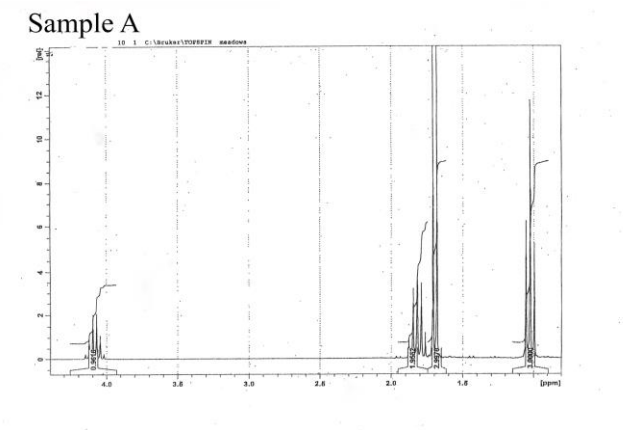
FTIR- The Fourier Transform Infrared Spectrometer (FTIR) stands out as a state-of-the-art instrument in infrared spectroscopy, offering advanced capabilities for compound analysis. Infrared spectroscopy, facilitated by FTIR, primarily serves the purpose of detecting functional groups within a compound. This is achieved by exposing the compound to infrared radiation and observing the resulting bending and stretching of its bonds. In the FTIR process, each functional group exhibits distinctive bending and stretching patterns, absorbing specific frequencies of the applied infrared radiation. No two compounds are known to have identical IR spectra. However, it's important to note that FTIR alone may not provide a comprehensive determination of a compound, as the results may lack sufficient information regarding the compound's backbone or parent chain. Instead, FTIR excels at identifying and characterizing the functional groups present. Utilizing an interferometer, FTIR measures the energy transmitted to the compound,

while a detector observes the amount of energy absorbed by the compound during the process. This technique allows for a detailed analysis of the functional groups within a compound, contributing valuable insights into its molecular composition.

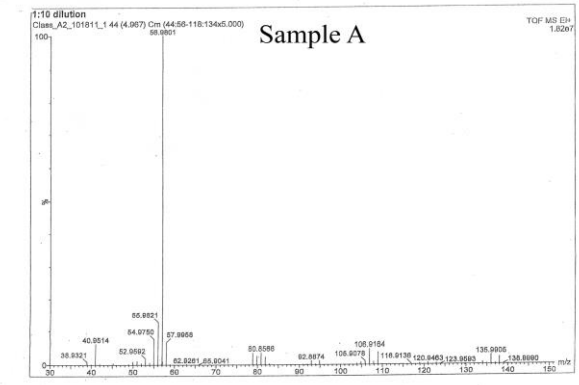
MS- The primary purpose of mass spectrometry is to attain the mass of a compound. Frequently, it is employed to identify compounds based on their overall molar weight. By comparing the data obtained from the mass spectrometer with the known molar weights of compounds, the identity of a substance can be deduced. A typical mass spectrometer comprises three integral components: an ionization source, a mass analyzer, and an ion detection system. The process begins with the ionization source, converting molecules into gas-phase ions, making them amenable to manipulation by external magnetic and electromagnetic fields. Once ionized, the molecules undergo sorting based on their mass-to-charge ratios through a mass analyzer. The separated ions are then measured, and the resulting data is transmitted to a data system for analysis and graph creation. In a mass spectrum, each peak signifies a distinct mass-to-charge ratio present in the sample. The heights of these peaks indicate the abundance of the corresponding ratio in the compound, offering valuable insights into the composition of the substance under examination.

Data

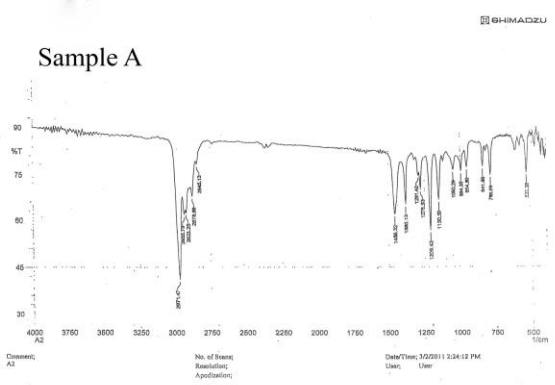
NMR



GCMS



FTIR



Analysis

NMR- When addressing the first NMR spectra reading the indication of four individual excitement areas indicates that there are four groups of protons present in the spectra. The integration values can be noticed next representing the relative number of each kind of proton in the given molecule. The integration values read 3.0000, 2.9970, 1.9557, 0.9616 from right to left. The integrals for this spectrum can be simplified to 3, 3, 2, 1. On the far right of the spectrum reading, the presence of a doublet can be seen. This also indicates that one of the carbons presents an unknown n, possibly a halide group. This reading indicates that there are two primary carbon atoms, one secondary, and one tertiary carbon atom. A skeleton of a structure similar to the unknown molecule can be drawn just from this data alone, although further action must be taken for further confirmation.

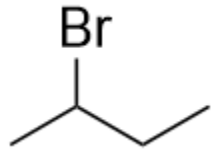
FTIR- The first step when addressing the IR spectrum is checking the 1850-1630 1/cm range. There are no dominant sharp peaks here so the presence of a C=O group is unlikely. Moving on to the 3550-3200 1/cm range, the presence of a strong, broad signal here would indicate the hydroxyl group of an alcohol. This is not present so this can be ruled out. From the IR spectrum reading provided the first function group indicated by the largest peak falls into the alkane (C-H) functional group region indicated by a frequency of 2971.47. There are no other dominant peaks in the IR spectrum that would indicate the presence of a C≡N between 2280-2100 1/cm range.

MS- The GCMS sample given for unknown compound A has its base peak at 57 m/z which is completely expected. Our four-carbon chain, as mentioned earlier, with nine hydrogen present makes for further confirmation. Four carbons at 12 amu is equal to approximately 48 m/z. When you add the additional nine hydrogen atoms at 1 amu this gives you the base reading of 57 m/z. The information provided here gives a reason why the base peak would be indicated at 56.9801

m/z. Another peak is indicated on the far right of the spectrum by a straight upward line at 135.9905 m/z. When you take this reading and subtract the base peak you are left with about 78 m/z which would indicate the loss of a bromine atom. This reading gives further confirmation of the suspicion of a halide group from the NMR reading.

Conclusion

Based on the analysis of the three provided spectra for the identification of unknown A, the most fitting compound appears to be 2-Bromobutane. The NMR spectrum provided insight into a structure comprising four carbons and nine hydrogens, suggesting the presence of a halide group. The FTIR spectrum supported this by indicating the absence of amine or alcohol groups and ruling out the possibility of oxygen. The final confirmation came from the GCMS reading, which aligned with the suspicions raised by the NMR readings. In conclusion, the comprehensive analysis of the NMR, FTIR, and GCMS spectra collectively points to the identification of the unknown compound A as 2-Bromobutane. A simplified bond line drawing is provided below for reference.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.